

GPU-Accelerated Continuous Subgraph Matching

Why BFS Beats DFS on the SIMT Architecture

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Amazon (AZ) 8v

LiveJournal

Continuous Subgraph Matching (CSM)

$$G \triangleq (v_1, v_2, \ell) \ q$$

- - & — Bytedance BG3 [9]
 - — Twitter motif detection [1]
 - & [4]
- **CPU** GraphFlow [2], TurboFlux [3], SymBi [7], CaLiG [8], NewSP [6]
- **ParaCOSM (ICPP'25)** [5] 5 CPU
 - Inner-update
 - Inter-updatesafe

→ GPU

- $g = (V, E) / L_V, L_E$
- (Def. 2.2) $M: V(Q) \rightarrow V(G)$ &
- (Def. 2.3) $\Delta \mathcal{G} = (\Delta G_1, \Delta G_2, \dots)$ $\Delta G = (\pm, e/v)$
- CSM (Def. 2.4) ΔG
 $\Delta \mathcal{M} = \mathcal{M}' \Delta \mathcal{M}$
- Compatible Set (Def. 2.5) $C(u, \phi) \text{ ---}$

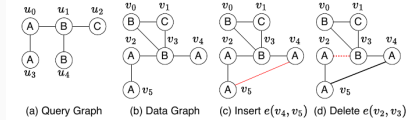


Figure 1: Running example of continuous subgraph matching

ParaCOSM [5] Fig. 1

Offline

- $\mathcal{A}$
- $\mathcal{O}$
- $\mathcal{M}_0$

Online ΔG

1. $+$ $\mathcal{A}$
2. $e \in \mathcal{O} \mathcal{T}$
3. $\Delta \mathcal{M}$

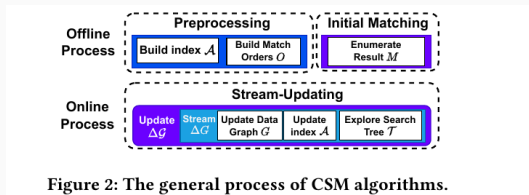


Figure 2: The general process of CSM algorithms.

ParaCOSM [5] Fig. 2

ParaCOSM $\geq 98.4\%$ safe/unsafe label / degree / candidate
unsafe $\rightarrow$ GPU

A DFS

CPU

- 
-  Hub
- 

B BFS

-  Warp lockstep
-  Prefix-scan
- 

(algorithm, query) **BFS** **DFS** GPU `gpu_bfs_versioned`

GPU ParaCOSM

- `gpu_classifier` `-m gpu`
safe / unsafe
- `gpu_candidate_filter` PFM2
query candidate
- `gpu_search` `-m gpu_all`
DFS
- `gpu_bfs_search` `-m gpu_bfs[_versioned]`
BFSpartial-match device

Versioned unsafe BFS

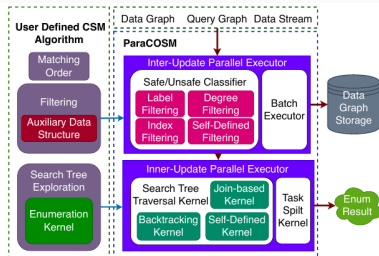


Figure 5: PARACOSM architecture.

ParaCOSM [5] Fig. 5GPU Inter-/Inner-Update Executor

Versioned BFS

1. unsafe (v_1, v_2) frontier
2. partial-match `ver_id`
3. query join + filter
4. prefix-scan coalesced partial-match buffer
5. Buffer $\rightarrow$ flush

/ / CSR BFS batch

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- A100-SXM4-80GB $\times$ 1
- Xeon Platinum 8470, 96
- 1.7 TB DRAM
- oneAPI 2025.3 (icpx)
- CUDA 12.9, sm_80

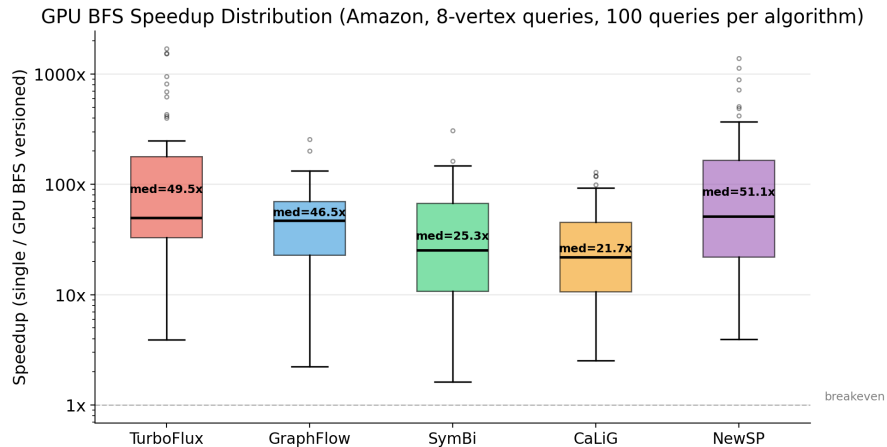
&

- $AZ|V| \approx 0.4M$, $|E| \approx 3.4M$
- $LJ|V| \approx 4.85M$, $|E| \approx 38.6M$
- 100 8-
- 1800s timeout / pair

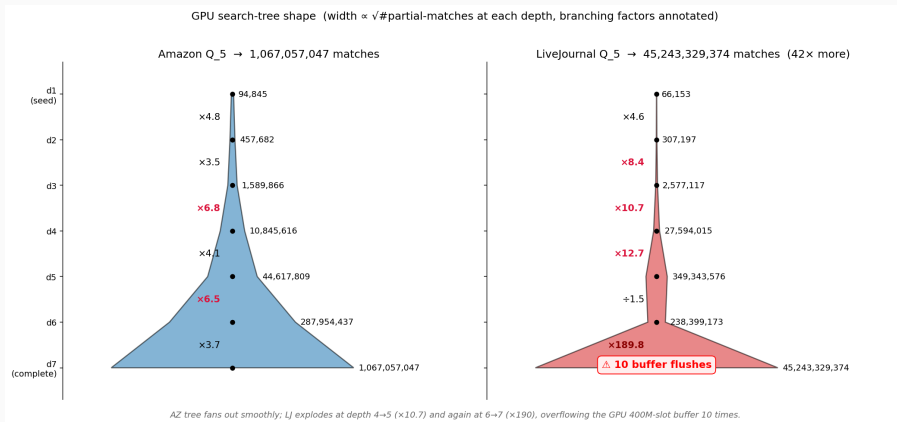
CPU -m single -t 1

PARALLEL_NEWSP	51.1×
PARALLEL_TURBOFLUX	49.5×
PARALLEL_GRAPHFLOW	46.5×
PARALLEL_SYMBI	25.3×
PARALLEL_CaLiG	21.7×

CaLiG CPU candidate-set CPU PCIe + kernel-launch



BFS — AZ Q_5 (GraphFlow)



(depth 4–5)
partial-match 4×10^8 bufferBFS flush

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LJ 8vpicture turns mixed

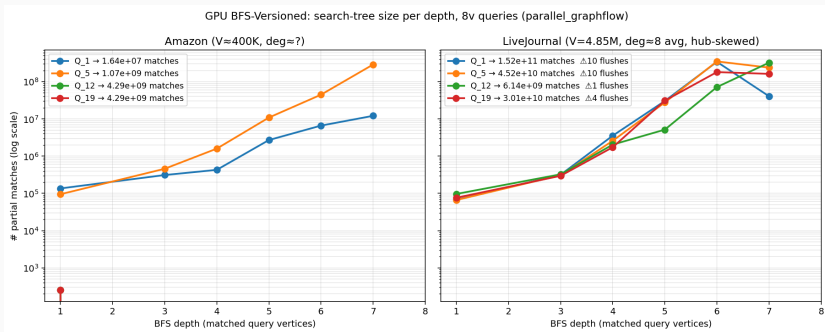
	pairs	cpu (s)	gpu (s)	
GRAPHFLOW	1	1494.6	105.4	14.18×
TURBOFLUX	9	666.3	215.2	3.10×
CALiG	14	583.6	658.9	0.89×
NEWSP*	16	87.1	637.0	0.14×
SYMBI	0	—	—	—

*NEWSP CPU single 0 adapter stub—

NewSP CPU stub bug

```
size_t Parallel_NewSP_Adapter::EnumerateNewEdge(  
    uint v1, uint v2, uint label, size_t /*tid*/) {  
    // Not implemented for NewSP adapter -- use versioned mode  
    return 0;  
}
```

- CPU single matching → CPU
- 4 GPU CPU 7 → BFS
- CPU -m versioned



LJ AZ

LJ NEWSP Q_197.97 $\times 10^{10}$ partial-match buffer flush 15

```
[BFS-V] Depth 5->6: 4.83e7 -> 3.85e8  
[BFS-V] Flush 400000000 partials at depth 6->7 (x15)  
Search complete: 7.97e10 matches, 1019.0 s
```

device buffer flush BFS stall

- **Edge-centric**
binary-search row-split hub
- **Partial-match streaming**
buffer D2H pinned memory BFS
→ LJ flush thrashing
- **H2D / search overlap**
stream BFS

- **DFS $\leftrightarrow$ BFS**
BFS $\rightarrow$ DFS
query + BFS
- **Early intersection pruning**
partial-match backward-neighbor
BFS LJ

- **GPU**

- Query GPU query
- Update safe-update batch GPU

- **CPU / GPU**

- Safe update $\rightarrow$ CPUunsafe $\rightarrow$ GPU
- NEWSP CPU versioned
- data-driven

Take-aways

1. GPU CSM **BFS** DFS
2. AZ 8vBFS-versioned 5 SOTA $21\times$ — $51\times$
3. LJ device buffer H2D flush
4. partial-match profiling
GPU graph mining workload

ParaCOSM, branch feat/gpu-profiling-charts
& docs/gpu_csm_report/

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